Colorectal cancer and diet in an Asian population--a case-control study among Singapore Chinese.

A hospital-based case-control study of diet and colorectal cancer was conducted among Chinese in Singapore (who constitute 77% of the population). A total of 203 cases and 425 controls were included. A history of the usual dietary intake one year prior to interview was taken using a quantitative food frequency questionnaire. Daily intakes of nutrients and selected food items were computed and stratified by tertiles of the control range, to assess risk in low-, medium- and high-intake categories. Effects were adjusted in analysis for age, sex, Chinese dialect group and occupation. For cancers of colon and rectum combined, significant observations were a protective effect of high cruciferous vegetable intake (OR = 0.50, p less than 0.01) and a predisposing effect of a high meat/vegetable consumption ratio (OR = 1.77, p less than 0.05). Similar results were observed for colon cancer alone. For rectal cancer alone (only 71 cases), significant (p less than 0.05) protective effects were observed for high intakes of protein (OR = 0.61), fibre (OR = 0.46), beta-carotene (OR = 0.54), cruciferous vegetables (OR = 0.51) and total vegetables (OR = 0.51). When further assessed by multiple logistic regression, tests for trend and assessment of risk in the extreme highest and lowest quintiles of the control range, the factors consistently significant were cruciferous vegetable intake and the meat/vegetable ratio. A particularly high relative risk was also noted in association with low coffee consumption (OR = 1.59, with p less than 0.05 for trend). No consistent trends were noted for fat or fibre intakes. For non-dietary variables investigated, a history of cholecystectomy increased the risk of both cancers combined (OR = 3.43, p less than 0.05) and colon cancer alone (OR = 4.39, p less than 0.01). This study in an Asian population of countries of Southern and Eastern Asia newly undergoing industrialization and in which rapid economic change is reflected in changing cancer patterns, suggests that the protective effects of certain dietary constituents, notably the cruciferous vegetables, may be more important than the hitherto stressed carcinogenic potential of fat and protein.